A linear decrease in magnetic field and temperature as a function of time (

Title
Names
Results
Introduction And motivation
Physical system
Methods used
Discussion and conclusion
Literature citation, resources, and acknowledgements

Background:
The earth is constantly being bombarded by radiation from the sun. The earth's magnetic field, or *magnetosphere*, protects us from these particles by deflecting them away (figure 2). Interactions between the sun and the earth's magnetic field result in interesting phenomena that are not fully understood. For instance, the aurora borealis occur due to solar particles accelerated by the earth's magnetic field at its north pole entering the atmosphere and colliding with nitrogen and oxygen atoms. These interactions are also observed on other planets in our solar system (figure 3)! Moreover, the sun's magnetic field is strong enough compared to earth's that it can compress the magnetic field lines on the side that faces the sun (the day side), and stretch them out on the night side.

Our sun is approximately 400 times larger than our moon, and 400 times as far away. Because of this, when the moon is aligned perfectly, it can completely block the sun in what we know as a *total solar eclipse* (figure 1). This offers an interesting opportunity to observe electromagnetic phenomena while the sun is completely blocked. We sought to ask: how is the earth's magnetic field impacted by the moon fully eclipsing the sun during a total solar eclipse?

Methods:
PNIRM3100 magnetometer (same model used on NASA's Artemis mission!) and MCP9808 temperature sensor were used to collect magnetic field and temperature readings during the total solar eclipse on 4/8/2024 in Burton, Ohio (figure 4).

Results:

Discussion:

In a strong magnetic field, field lines are observed very close to one another. Likewise, field lines that are farther apart indicate a weaker field. If when the sun is facing one side of the earth it can compress the magnetic field lines, it would be reasonable to suggest that the magnetic field of the earth increases when it interacts with the sun. Therefore, when the moon is blocking the sun, the magnetic field activity would decrease. This was observed in our EZIE-Mag measurements, where there was a steady decrease as the sun became progressively more eclipsed by the moon, reaching its lowest during totality.

Supplementary figure 1: Solar corona during t

As the sun became progressively more eclipsed by the moon, a steady decrease in magnetic field measurements was observed.

A steady decrease in magnetic field was observed as the sun became progressively more eclipsed by the moon and eventually became totally covered.

The sun's magnetic field is known to compress the earth's on the day side and stretch it on the night side. In a strong magnetic field, field lines are observed very close to one another. It would therefore be reasonable to suggest that the local magnetic field strength of the earth increases when it interacts with the sun. When the moon is blocking the sun, the magnetic field activity would therefore decrease. This was observed in our EZIE-Mag measurements, where there was a steady decrease as the sun became progressively more eclipsed by the moon, reaching its lowest during totality.

https://earth-planets-space.springeropen.com/counter/pdf/10.1186/s40623-022-01656-9.pdf
https://www.popularmechanics.com/science/environment/a26344034/sun-magnetic-field-aurora/
https://scied.ucar.edu/learning-zone/sun-space-weather/earth-magnetosphere
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Methods:

PNIRM3100 magnetometer (model used on NASA's Artemis mission) and MCP9808 temperature sensor were used to collect magnetic field and temperature readings during the total solar eclipse on 4/8/2024 from Burton, Ohio.

When these solar particles interact with the earth's

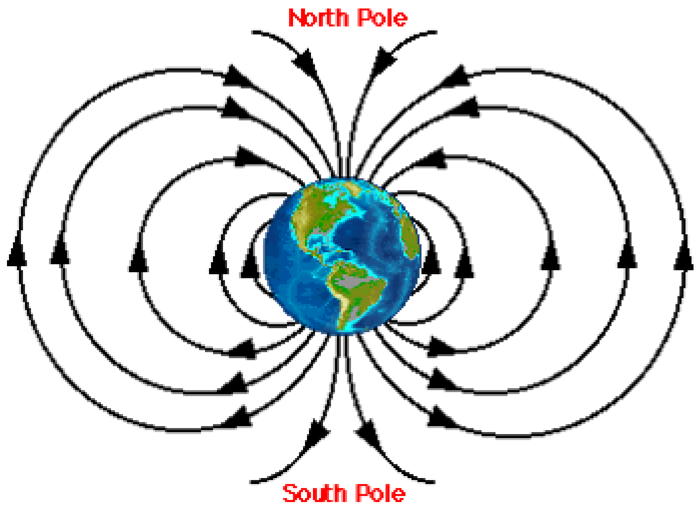
or the sun's magnetic field interact with the earth's magnetic field, interesting phenomena happen.

Due to this when our moon is aligned perfectly, a total solar eclipse can be witnessed

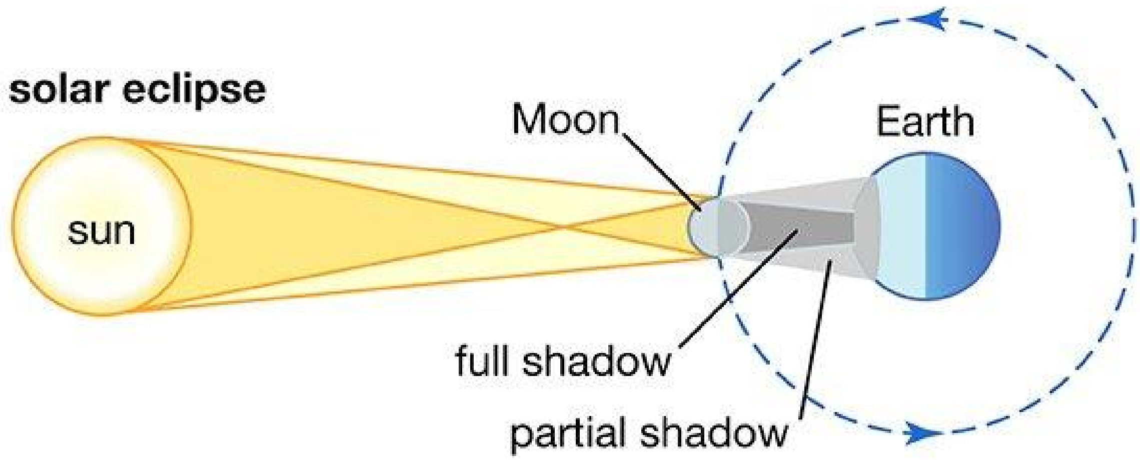
During a solar eclipse, the moon comes between

Results:
Temperature decrease leading up to totality and drastic decrease during totality.

Figure 1: Auroras on earth and other celestial bodies
Figure 2: Earth's magnetic field
Figure 3: Visualization of a total solar eclipse


North Pole
South Pole


Earth
Saturn
Jupiter
Brown Dwarf LSR J1835+3259
AURORA
Credit: NASA/ESA
Hubble/Caltech


solar eclipse
sun
Moon
Earth
full shadow
partial shadow